

Literature-Implied Status Note for arXiv:1305.6288

Run `fa.banach.001`

July 18, 2026

Status. `literature_implied_answer` (partial subcase).

Source Question

The source paper is Tomasz Kobos, *Equilateral dimension of some classes of normed spaces*, arXiv:1305.6288v2. In the concluding remarks, after the codimension-one result for subspaces of ℓ_∞^n , Kobos asks:

Prove that for any $(n - 2)$ -dimensional subspace X of ℓ_∞^n we have $e(X) \geq n - 1$.

Here $e(X)$ denotes the maximum cardinality of an equilateral subset of X .

Supporting Literature

The supporting paper is Nora Frankl, *Large Equilateral Sets in Subspaces of ℓ_∞^n of Small Codimension*, Discrete Comput. Geom. 67 (2022), 882–893, DOI 10.1007/s00454-020-00272-2.

Frankl explicitly identifies Kobos' problem in the introduction and states that Theorem 1.2 solves it in large ambient dimension. The relevant theorem says that if X is an $(n - k)$ -dimensional subspace of ℓ_∞^n , then, among other lower bounds,

$$e(X) \geq 1 + \frac{1}{2^{k-1}} \sum_{r=1}^{\ell} \binom{n - k\ell}{r} \quad \left(1 \leq \ell \leq \frac{n}{k+1}\right).$$

Taking $k = 2$ and $\ell = 2$ gives

$$e(X) \geq 1 + \frac{1}{2} \left[\binom{n-4}{1} + \binom{n-4}{2} \right].$$

For every $n \geq 9$ this lower bound is at least $n - 1$. Thus Frankl's theorem implies Kobos' desired inequality for codimension-two subspaces of ℓ_∞^n whenever the ambient dimension satisfies $n \geq 9$.

Scope

This packet records a literature-implied partial answer, not a new proof from the run. The implication is direct but requires matching notation: Frankl's ambient dimension n is the same ambient dimension as in Kobos' final problem, and $k = 2$ is the codimension.

The source problem is not fully closed by this packet. The cases $n \leq 6$ follow from the known validity of Petty's conjecture in dimensions at most 4, since then $\dim X = n - 2 \leq 4$. Frankl covers $n \geq 9$. The ambient dimensions $n = 7$ and $n = 8$ remain outside this identification.

Search Evidence

The run indexes had no prior packet for arXiv:1305.6288. Nearby equilateral entries concerned infinite equilateral sets, nonseparable examples, or other finite-dimensional questions. Web and local searches for the codimension-two subspace problem found Frankl's later open-access paper, whose introduction explicitly cites Kobos and says the theorem solves the problem for $n \geq 9$.

References

- [1] T. Kobos, *Equilateral dimension of some classes of normed spaces*, Numer. Funct. Anal. Optim. 35 (2014), 1340–1358. arXiv:1305.6288.
- [2] N. Frankl, *Large Equilateral Sets in Subspaces of ℓ_∞^n of Small Codimension*, Discrete Comput. Geom. 67 (2022), 882–893. DOI: 10.1007/s00454-020-00272-2.